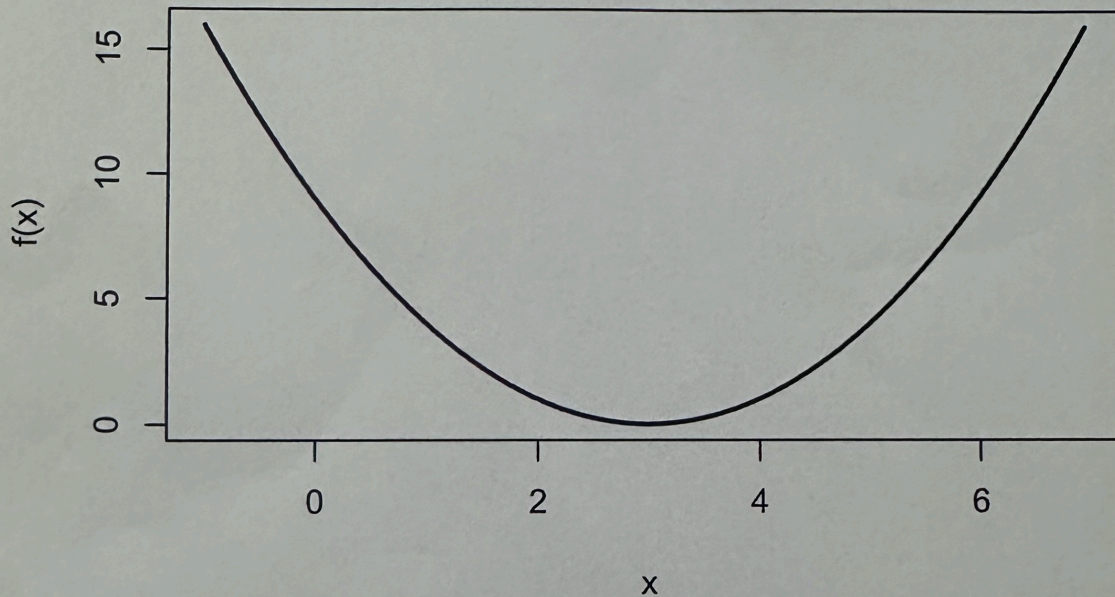


Solution:-

MTH209: Assignment 9

Consider minimizing the function $f(x) = (x - 3)^2$.



Starting from 0 and 4 as the two values of the vertices, write down three steps of the Nelder-Mead algorithm:

1. Step 1: $\theta_1 = 4$, $\theta_2 = 0$

2. Step 2: $\theta_1 = 4$, $\theta_2 = 2$ or $\theta_1 = 2$, $\theta_2 = 4$

3. Step 3: $\theta_1 = 3$, $\theta_2 = 4$ or $\theta_1 = 3$, $\theta_2 = 2$

4. Step $\theta_1 = 3$, $\theta_2 = 2.5$ or $\theta_1 = 3$, $\theta_2 = 3.5$

ROUGH WORK